What we have done (Team 4):

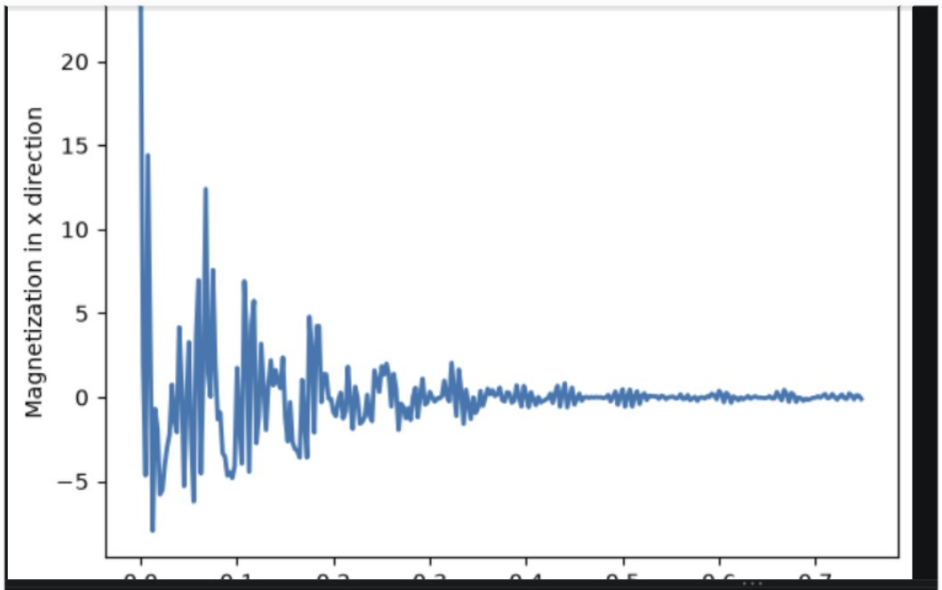
1) Matching the Molecules to the given Spectra (page 2-3)
2) Simulation on real quantum hardware (page 4)
3) Estimating the maximum tolerable noise strength (page 5)

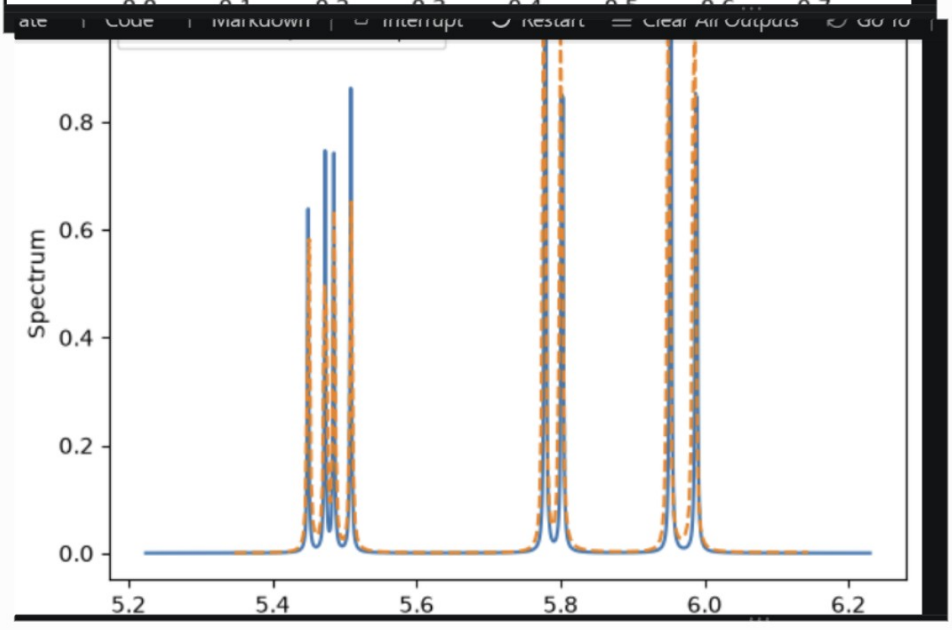
Team 4 (Correlated Qubits):

Jonas Höpker
Paul Seyffer
Henrik Holbein
Anubav Dasgupta
Marco Mönnich

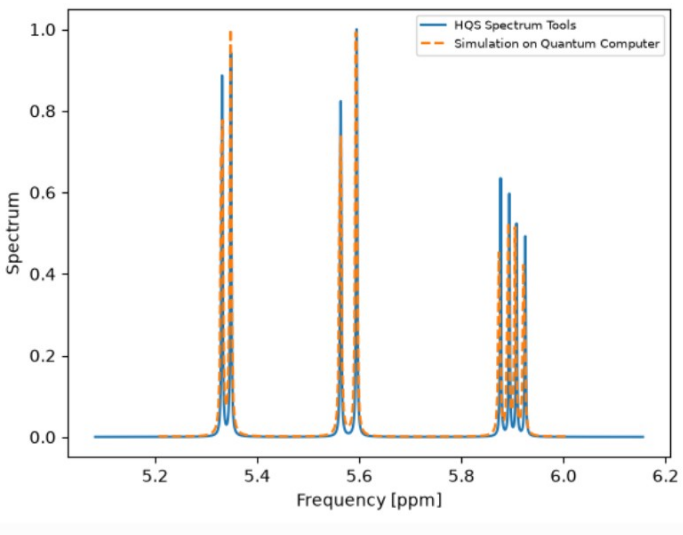
Molecule identification

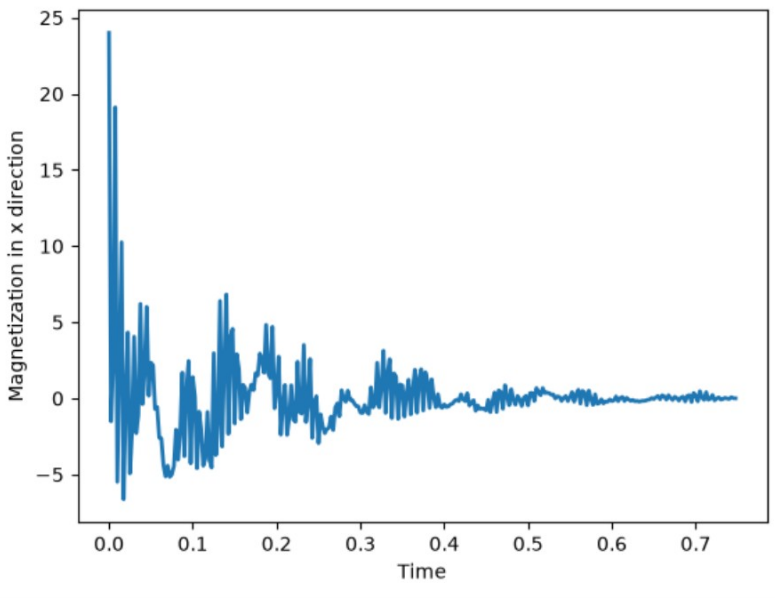
Spektrum 3 -> Molekül A



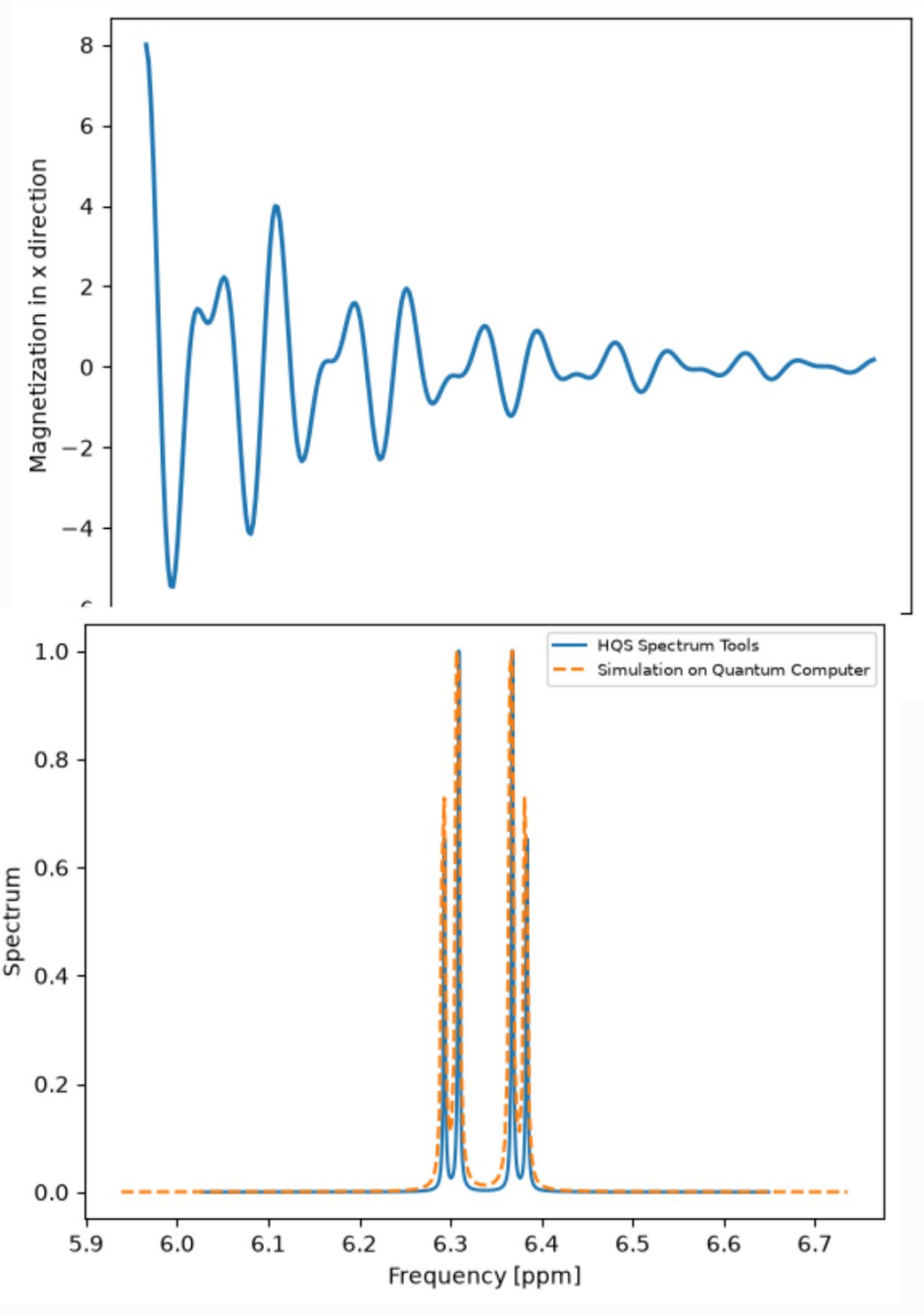


Spektrum 2 -> molekül B





Molecole C -> Spectrum 1



Classification:

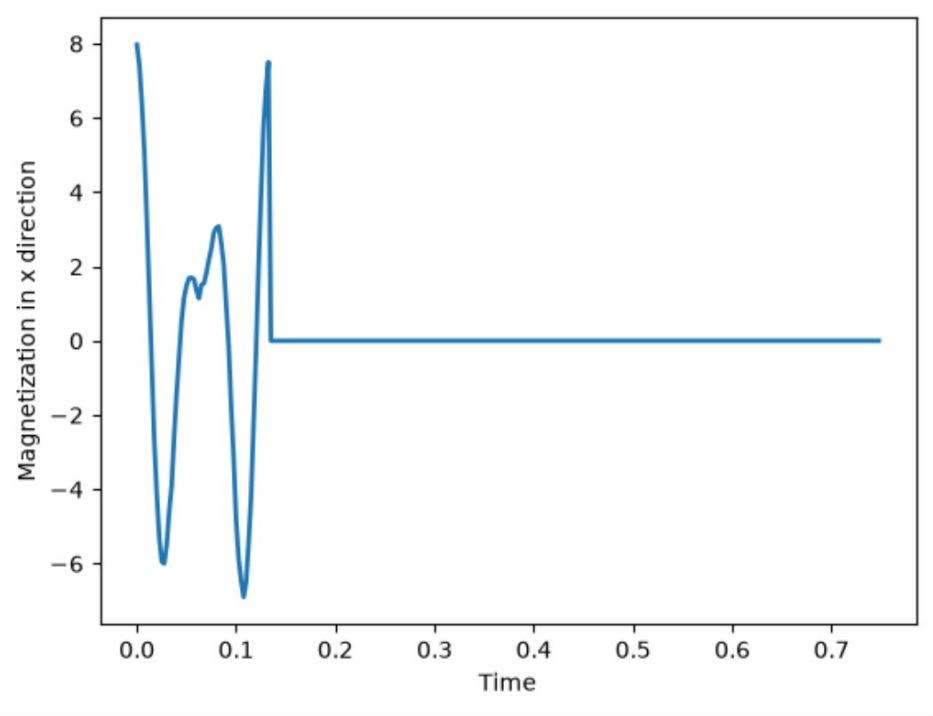
A: 3

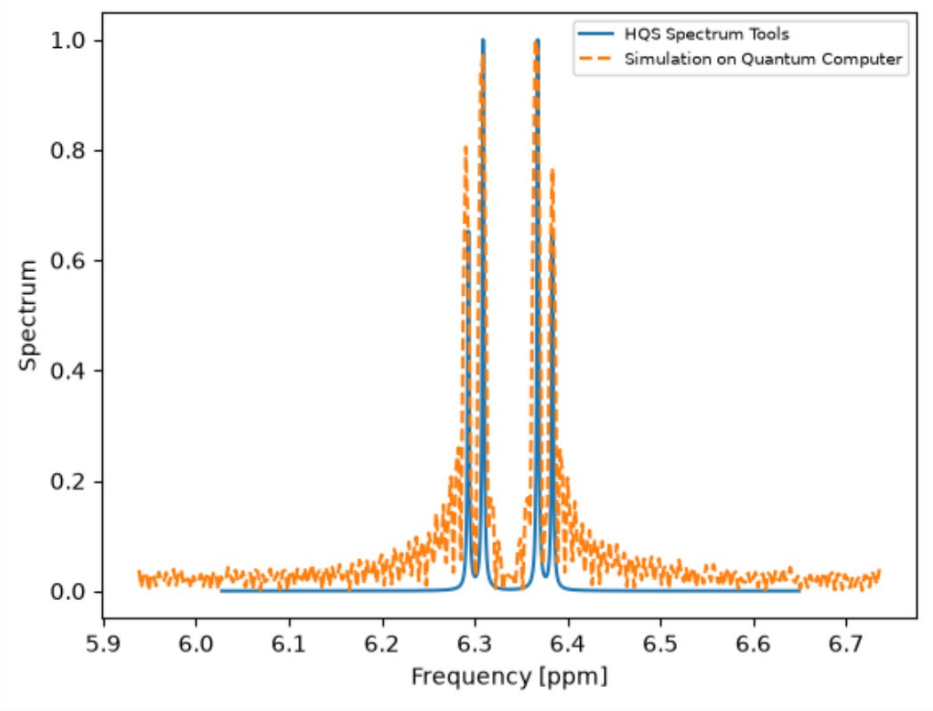
B: 2

C: 1

Run on a real hardware

Backend: ibm_fez, 156 qubits, native gates: ['cz', 'delay', 'id', 'if_else', 'measure', 'measure_2', 'measure_reset', 'measure_reset_2', 'reset', 'reset_2', 'rz', 'sx', 'x', 'xslow'] - "molecule C"





maximum allowed Noise, where all peaks are still clearly discernable:

Molecule A: 25
Molecule B: 26
Molecule C: 26

(for the plots also see the github repository https://github.com/jona-sh/qiskit-fall-fest)

